

Chapter 18

Production of Titanium and Titanium Alloy Powders

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INTRODUCTION

Titanium possesses an exclusive combination of low density, good mechanical properties at room and elevated temperatures, and good corrosion resistance. Titanium and its alloys are used extensively in the aerospace and biomedical industries as well as in other industrial applications. However, titanium ingots are expensive to produce and fabricate. Another problem is connected with segregation in highly alloyed titanium materials. The high cooling rate that PM technology makes possible ensures that significantly greater amounts of alloying elements remain in the solid solution compared with ingot metallurgy, and thus can substantially influence material phase composition and properties [1,2]. At the same time, PM processing permits articles that are close to their final shape. Therefore, this leads to reduced material loss during production as well as lower production costs.

There are many known processes of titanium and titanium alloy powder manufacture. Currently, chemical reduction, hydrogenation/dehydrogenation, gas atomization, plasma-rotating electrode, and plasma atomization processes are mainly used for commercial production. Small amounts of titanium powder are produced by electrolysis of titanium compounds (TiCl_4 , TiO_2) in melts, centrifugal atomization of the melt, amalgam metallurgy, and the disproportionation method of low titanium halogenides.

Development of new titanium production methods, such as the ITP/Armstrong, CSIRO (Commonwealth Scientific and Industrial Research Organization, Australia), and MER processes, is aimed at lowering the cost of PM titanium powder. However, these powders are not yet a commercial application.

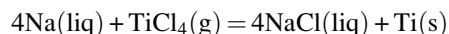
CHEMICAL REDUCTION

Generally, the initial source of titanium powder is “sponge fines,” so called owing to their appearance. Sponge fines are nodular particles (see Introduction, Fig. 2) of pure titanium produced during the chemical reduction process from titanium tetrachloride. The commercial manufacture of titanium metal includes the chlorination of natural or synthetically produced rutile, TiO_2 [1,3]. The obtained titanium tetrachloride is reduced to metallic titanium by the sodium reduction method known as the Hunter process, the magnesium reduction method known as the Kroll process, and the calcium hydride reduction process used in Russia.

In reduction processes of titanium tetrachloride, most of the residual chloride is removed by vacuum distillation or water leaching. The final sponge fines typically contain from 0.12 to 0.15 wt% Cl.

Sodium Reduction Method (Hunter Process)

This process invented in 1910 by Matthew A. Hunter was the first industrial process to produce pure ductile metallic titanium. Titanium tetrachloride is reduced with sodium to form pure metallic titanium (99.9%) by heating titanium tetrachloride with sodium at 973–1073 K:



Characteristic sponge fines manufactured by the Hunter process are shown in Fig. 18.1. A typical chemical analysis of commercial powders is given in Table 18.1. A screen analysis of sponge fines produced by sodium reduction is characterized by the content of the particle size fractions smaller than 105 μm in the amount of 78 wt% and fractions smaller than 64 μm and amount of 40.8 wt%.

Table 18.2 illustrates typical technological characteristics of titanium powders and tensile properties of Ti-6Al-4V compacts produced from sponge fines and an aluminum-vanadium master alloy.

Powder dispersivity grows at TiCl_4 bubbling through the melt, on the surface of which metallic sodium is supplied: the grain size of the basic mass of powder is lower than 20 μm . More finely dispersed powders are produced by low-temperature reduction of titanium's compounds in organic media by using alkaline metals or amalgams as reducing agents.

Though the Hunter process successfully produces very high-quality titanium, it was replaced by the more economical Kroll process. Currently, only a small facility uses the Hunter process to manufacture titanium sponge for high-purity sputtering targets [2].

Armstrong Process

Developed by Cristal Metals, formerly International Titanium Powders (ITP), the Armstrong process produces titanium powder by reduction of titanium tetrachloride with sodium, as the Hunter process does. However, this process is continuous in contrast to the batch mode of the Hunter process. The process begins with titanium tetrachloride that is reacted to form solid titanium powder and sodium chloride. The reaction is continuous and takes place in the titanium reactor.

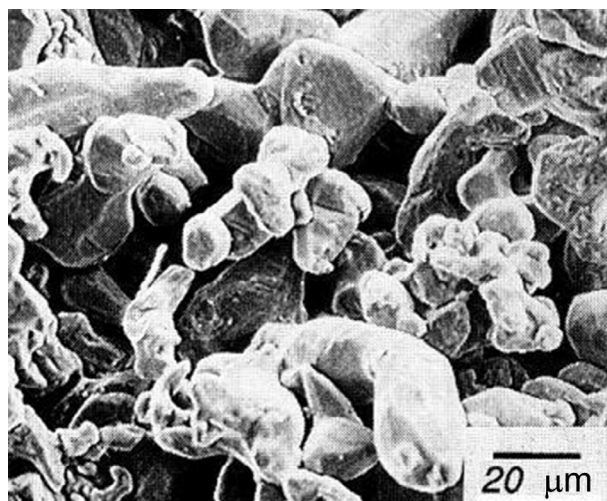


FIG. 18.1 Sponge fines produced by the sodium (Hunter) process.

TABLE 18.1 Typical Chemical Analysis of Chemical Reduction Process Titanium Sponge Fines, and Sinter-Compacted Ti-6Al-4V Produced From These Fines and Al-V Master Alloy Powders

Elements (Content, wt%)	Sodium Process			Calcium Hydride Reduction Process			Hydride/Dehydride Process		Electrolytic Process
	CP ^a	^b	CP ^c	CP ^a	^d	CP ^e	P6/4-3 ^f	CP ^g	CP ^g
Al	...	6.2	...	...	4.65	...	5.09	...	...
V	...	4.1	...	...	3.80	...	4.17	...	...
O	0.13	0.24	0.05–0.30	0.19	0.20	0.25–0.30	0.131	0.15–0.30	0.05–0.30
N	0.03	0.016	0.02–0.03	0.06	0.06	0.05–0.06	0.006	0.03–0.05	0.02–0.03
H	0.07	0.002	0.02–0.03	0.34	0.30	0.20–0.40	0.0015	0.06–0.15	0.02–0.04
C	0.02	0.02	0.01 max	0.03	...	0.04–0.05	0.02	0.015–0.020	<0.02
Fe	0.02	0.18	0.01–0.02	0.11	...	0.15–0.20	0.1	0.07–0.10	0.02–0.08
Na	0.10	0.10	...	...	...	...	...	...	...
Cl	0.13	0.12	0.10–0.13	0.004	0.003	0.003–0.005	...	0.06–0.08	0.03–0.08
Si	...	...	0.006 max	0.05	...	0.06–0.07	0.02	0.01–0.02	<0.02
Ca	...	...	...	0.04	0.06	0.08	...	...	...
Ni	...	...	...	0.07	...	...	...	...	...
Cu	...	...	...	...	...	...	<0.01	...	...
Sn	...	...	...	...	...	...	0.01	...	...
Nb	...	...	...	...	...	...	1.17	...	...
Mo	...	...	...	...	...	...	0.01	...	...
W	...	...	...	...	...	...	0.01	...	...
Cr	...	...	...	...	...	...	0.02	...	...
Ti	Balance	Balance	Balance	Balance	Balance	Balance	Balance	Balance	Balance

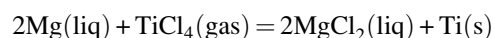
^aSource: Ref. [2].^bSinter compacted Ti-6Al-4V produced from these sponge fines and Al-V master alloy powder, source: Ref. [2].^c–630 +180 μm powder sizes, source: Ref. [1].^dSinter compacted Ti-6Al-4V produced from these fines, source: Ref. [2].^e–100 +40 μm (25%), –40 μm (75%) powder sizes, source: Ref. [1].^fComposition of alloys synthesized from blends of CP and master alloy powders.^g–630 +180 μm powder sizes, source: Ref. [1].

CP, commercially pure titanium.

Liquid sodium is pumped through a cylindrical chamber containing a centerline second tube. The TiCl₄ vapor is injected into the sodium stream through this inner tube/nozzle. The reaction occurs immediately downstream from the nozzle. The titanium powder is carried out in the excess sodium stream. Titanium, sodium, and sodium chloride are separated by filtration, distillation, and washing. The powder produced has a purity level near that of commercially pure grade 1 titanium, including a chlorine content of <50 ppm. By simultaneous reduction of other metal chlorides, it is possible to produce alloy powders.

Magnesium Reduction Method (Kroll Process)

This process occurs at 1073–1123 K:



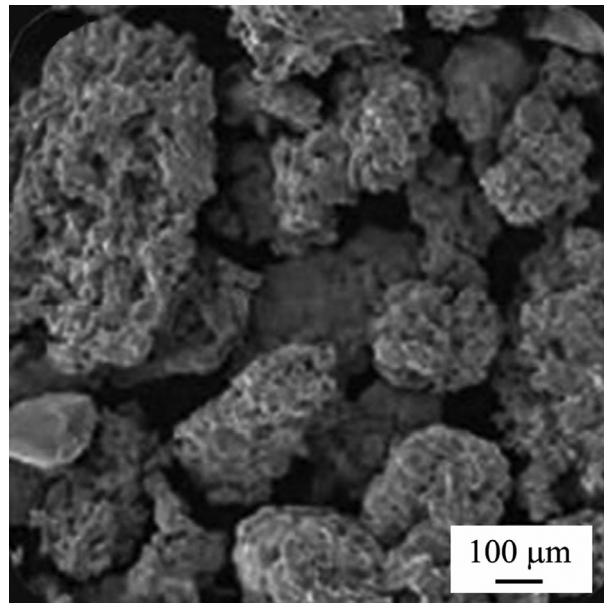
Removing impurities from the sponge is realized by vacuum distillation. However, the sponge (Fig. 18.2) is very strong and cannot be easily crushed to size to eliminate the trapped MgCl₂. The main drawback of the Kroll process is that the time

TABLE 18.2 Technological Characteristics of Titanium Powders and Sintered Compacts Produced From These Powders

Technological Characteristics	Sodium Process			Calcium Hydride Reduction Process			Hydride/Dehydride Process		Electrolytic Process
	CP ^a	^b	CP ^c	CP ^a	^d	CP ^e	P6/4-3 ^f	CP ^g	CP ^h
Tap density (g/cm ³)	...	...	0.9–1.2	...	...	0.95–1.40	...	1.35–1.70	1.4–1.8
Flow rate (s)	...	...	20 max	...	...	15 max	...	75 max	28 max
Yield strength (MPa)	...	868	...	...	865	...	868	...	...
Tensile ultimate strength (MPa)	...	945	...	...	966	...	965	...	...
Elongation (%)	...	15	...	...	13	...	14	...	...
Reduction of area (%)	...	25	...	...	32	...	31	...	...

^aSource: Ref. [2].^bSinter-compacted Ti-6Al-4V produced from these fines and Al-V master alloy powder, source: Ref. [2].^c–630 + 180 μm powder sizes, source: Ref. [1].^dSinter-compacted Ti-6Al-4V produced from these fines, source: Ref. [2].^e–100 + 40 μm (25%), –40 μm (75%) powder sizes, source: Ref. [1].^fComposition of alloys synthesized from blends of CP and master alloy powders.^g–630 + 180 μm powder sizes, source: Ref. [1].^h–180 μm powder size, source: Ref. [1].

CP, commercially pure titanium.

**FIG. 18.2** SEM photomicrograph of sponge made by the Kroll process.

required for purification of the titanium sponge by vacuum distillation can exceed 4–5 days, depending on the titanium sponge block size.

When the sponge is crushed to a required size, the riddlings (particles finer than 2 mm) are of 3%–5% of the mass of the sponge produced. They are used in powder metallurgy, in anticorrosive coatings, and for the production of titanium carbides, nitrides, and other compounds. The yield of powder fractions of magnesium-thermal titanium essentially increases if

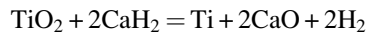
vacuum separation of the reaction mass by hydrometallurgical processing is used. However, the quality of powders in this case is impaired.

Titanium powders can be produced by continuous reduction of TiCl_4 with gaseous magnesium in a stream of argon. In this process, the titanium powder and magnesium chloride are continuously carried out of the reaction zone.

To produce titanium powder, mechanical crushing of the titanium sponge is done. However, in this case there are some difficulties connected with the plasticity of high-purity titanium, therefore the titanium is preliminarily oxidized or hydrogenated to embrittle it. Such a method is especially interesting for crushing titanium alloys, Ti-Al alloys in particular, produced by an aluminothermic reduction of titanium dioxide.

Calcium Hydride Reduction Process

The reduction of titanium dioxide with calcium hydride is carried out at temperatures of 1423–1473 K:



The reaction product is processed by a water solution of hydrochloric acid to remove CaO and Ca(OH)_2 , then the powder is washed in distilled water, rinsed with alcohol, and dried at 333–343 K in vacuum.

Powder particles produced by the calcium hydride process have a spongy morphology similar to sponge fines produced by the sodium process, and consist of conglomerations of small particles fused together during the reaction process. The powder produced (size fraction lower than $200\text{ }\mu\text{m}$) contains up to 90% of particles $<10\text{ }\mu\text{m}$ in size and has an apparent density of $0.6\text{--}0.8\text{ g/cm}^3$. The powder contains up to 0.05% of calcium as its oxide. The average particle size of commercially pure titanium powder is $41\text{ }\mu\text{m}$ [4].

Table 18.1 shows a typical chemical analysis of commercial powders produced by calcium hydride reductions. Table 18.2 gives typical technological characteristics of titanium powders and tensile properties of Ti-6Al-4V articles produced from calcium hydride-reduced titanium and aluminum-vanadium master alloy powders.

Hydrogenation/Dehydrogenation (HDH) Process

This method is based on the reaction of titanium with hydrogen at 623–973 K, the formation of fragile hydrides, and their mechanical crushing and subsequent dehydrogenation during 1–2 h at 973–1073 K to a powder-form product. A saturation of titanium by hydrogen achieves 2–3.5 wt%, depending on the purity of the initial material [1]. The hydrogenated titanium in possession of an angular shape (Fig. 18.3) is very brittle and can be simply ground to a fine powder. Mechanical crushing in ball mills allows manufacturing powders of the required grain size. The disadvantages of the method are difficulties of

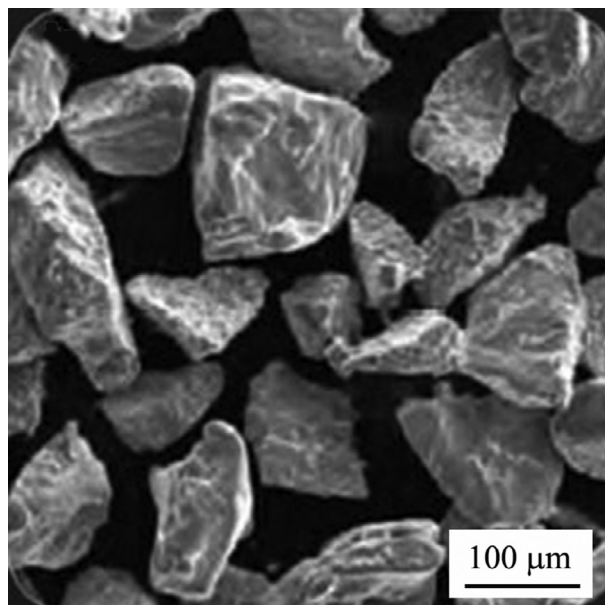


FIG. 18.3 SEM photomicrograph of angular hydrogenation/dehydrogenation (HDH) titanium powder.

hydrogen purification from oxygen and nitrogen as well as high sensitivity of titanium to the content of the residual hydrogen, causing fragility.

Typical chemical analysis and technological characteristics of titanium powders and sintered compacts are given in Tables 18.1 and 18.2.

Hydride of electrolytic titanium is distinguished by high purity and is used very effectively as a generator of hydrogen, in particular, in the annealing of watch fibers and as an additive to the charge of various metal powders that activates the process of sintering. Crushed titanium hydride can serve as a raw material in the production of materials such as titanium carbides, borides, nitrides, and silicides.

For reducing the titanium powder cost, the hydrogenation process followed by hydrometallurgical processing was suggested [5]. The processes of dehydration with complete or partial removal of hydrogen were developed.

Chemical reactions of continuous and semicontinuous processes for manufacturing titanium hydride using titanium chlorides of different valences are listed in Table 18.3 [6]. Because reactions 3 and 10 under normal conditions have low yield and require either a high vacuum or special conditions (rapid quenching), magnesium is used for titanium dichloride reduction in industrial production.

The hydrogenated titanium powder produced at ADMA Product Inc. (<https://www.admaproducts.com/>) was used to manufacture titanium alloy parts for various applications, including aircraft and military, by the blended elemental PM process. The Ti-6Al-4V bars were manufactured by cold isostatic pressing/sintering/rotary forging from TiH₂ powder [2].

PLASMA-ROTATING ELECTRODE PROCESS (PREP)

PREP is a centrifugal atomization process where the end of a consumable metal bar is melted while being rotated about its longitudinal axis. The molten metal is centrifugally atomized in the form of droplets that solidify in an inert gas atmosphere to spherical powder particles. The consumable electrode can be melted by various energy sources. For producing titanium-alloyed powders in the United States, helium plasma is used [3]. The electrodes are prealloyed bars of 60–65 mm in diameter and are rotated at speeds up to 15,000 rpm. Particle size depends on the alloy composition, the electrode diameter, and the rotation speed. Typical sizes for Ti-6Al-4V powders are between 100 and 300 μm with a median diameter of about 175 μm. The PREP powder is spherical (Fig. 18.4) while having good flow characteristics and a tap density about 65% of

TABLE 18.3 Chemical Reactions of Continuous and Semicontinuous Processes for Manufacturing Titanium Hydride Using Titanium Chlorides of Different Valency

No	Formula	Temperature (K)	Environment	Comments
1	$\text{TiO}_2 + 2\text{C} + 4\text{HCl} = \text{TiCl}_4 + 2\text{CO} + 2\text{H}_2$	1073–1273	Atmospheric	
2	$\text{TiCl}_4 + \text{Ti} = 2\text{TiCl}_2$	923–1000	Atmospheric	
3	$2\text{TiCl}_2 = \text{Ti} + \text{TiCl}_4$	1073	Atmospheric or vacuum	The reaction rate at atmosphere pressure
4	$2\text{TiCl}_4 + \text{H}_2 = 2\text{TiCl}_3 + 2\text{HCl}$	773–973	Atmospheric	
5	$\text{TiCl}_2 + \text{H}_2 = \text{Ti} + 2\text{HCl}$	1273	Atmospheric	Excess of hydrogen
6	$2\text{TiCl}_3 + \text{H}_2 = 2\text{TiCl}_2 + 2\text{HCl}$	773	Atmospheric	
7	$\text{TiCl}_4 + \text{H}_2 = \text{TiCl}_2 + 2\text{HCl}$	1072–1123	Atmospheric	Excess of hydrogen
8	$2\text{TiCl}_3 = \text{TiCl}_2 + \text{TiCl}_4$	723 773	Vacuum Atmospheric	
9	TiCl_4	1637	Atmospheric	Evaporation
10	$\text{TiCl}_4 + 2\text{H}_2 = \text{Ti} + 4\text{HCl}$	1273–2273	Atmospheric	Excess of hydrogen
11	$3\text{TiCl}_4 + \text{Ti} = 4\text{TiCl}_3$	923–1000	Atmospheric	
12	$\text{Ti} + \text{H}_2 = \text{TiH}_2$	673–1023	Atmospheric	

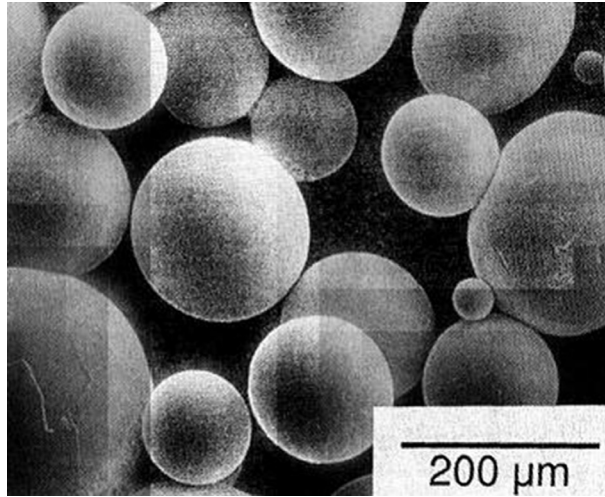


FIG. 18.4 Ti-6Al-4V powder produced by the plasma rotating electrode process.

solid substance. Various process modifications were investigated to produce finer powder [7]. In particular, the rotating electrode process (REP) was combined with gas atomization.

The rotating electrode process is also used by similar technologies in Russia, China, and Japan [1,3]. This process is discussed in more detail in Chapter 4. The PREP and REP powders are used to produce PM articles in aerospace, gas turbine engineering, environment control, and biomedical applications.

GAS ATOMIZATION

The plasma atomization process was developed to produce fine spherical titanium powder using titanium wire as the starting material [8]. In this process, the wire is fed into the apex of three plasma torches where it is melted and atomized in an argon atmosphere (Fig. 18.5). Droplets are then cooled during their flight in argon with a cooling rate in the range of 10^2 – 10^3 K/s. They are then solidified, forming spherical powder particles. Spherical titanium and Ti-6Al-4V alloy powders

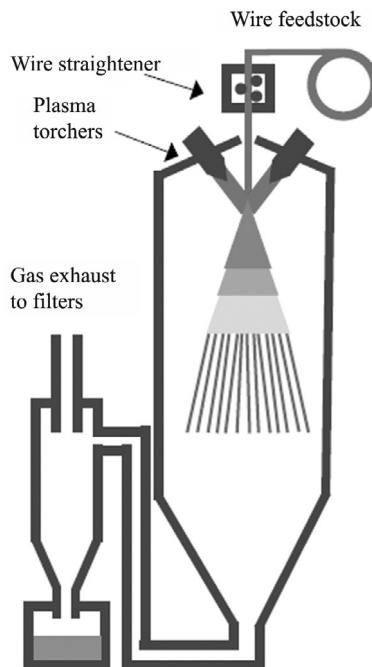


FIG. 18.5 Schematic of the plasma atomization process developed to produce fine spherical titanium powder from titanium wire.

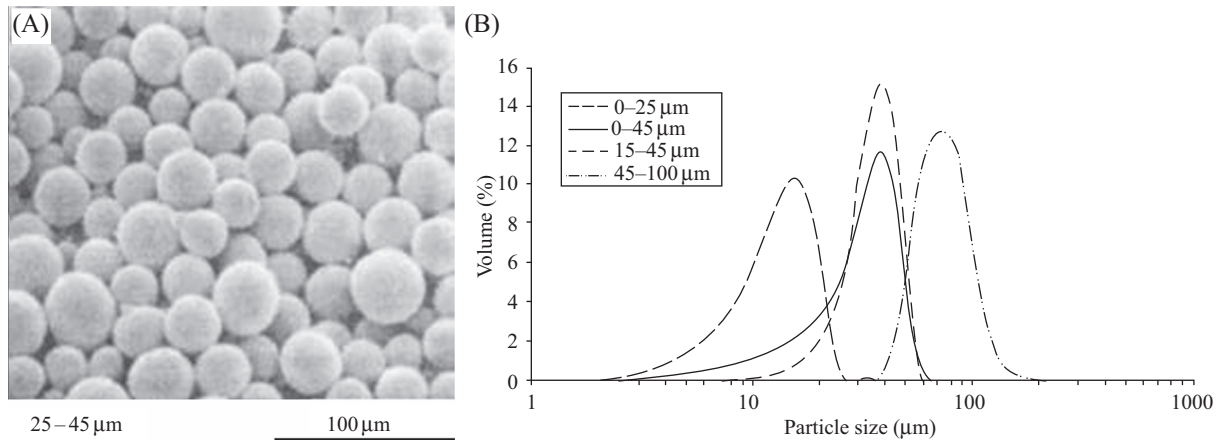


FIG. 18.6 Plasma-atomized powders (A) SEM photomicrograph of 25–45 μm size fraction, (B) Differential curves of particle size distribution of 4 size fraction.

with particle size from 0–25 to 0–180 μm have been commercialized in six size grades (Fig. 18.6) [9]. The oxygen content depends on particle size and is in the range 0.1–0.3 wt%. Apparent density depending on size fraction is in the 2.52–2.75 g/cm³ range. The flow rate varies depending on particle sizes in the ranges from 23 to 200 s/(50 g). PyroGenesis Inc. proposes the use of plasma-atomized titanium powders for producing porous filters, applications in injection and thermal spray processes, and biomedical applications [8].

Plasma melting induction guiding gas atomization (PIGA) is another variation for producing ceramic-free Ti alloy powders using a gas atomization process developed by GKSS Research Centre in Geesthacht, Germany [10]. In this process, a prealloyed rod is introduced into a conical induction coil where the tip of the rod is heated. Once the melting temperature is reached, the melt drops down into the center of a nozzle. The obtained powder particles are predominantly in a spherical shape. The resulting finer 45 μm powders are used in metal injection molding (MIM) and other areas. The Leybold firm (Germany) has developed a technology to produce spherical powders of titanium and its alloys by melt atomization by inert gas [11]. The metal is melted in induction or electron-beam furnaces. Atomization is performed by argon or on a rotating disc.

The gas-atomized prealloyed Ti-6Al-4V alloy has a characteristic spherical shape (Fig. 18.7). For the purpose of decreasing the cost of argon-atomized powder manufacture, a premixed raw material consisting of titanium sponge and

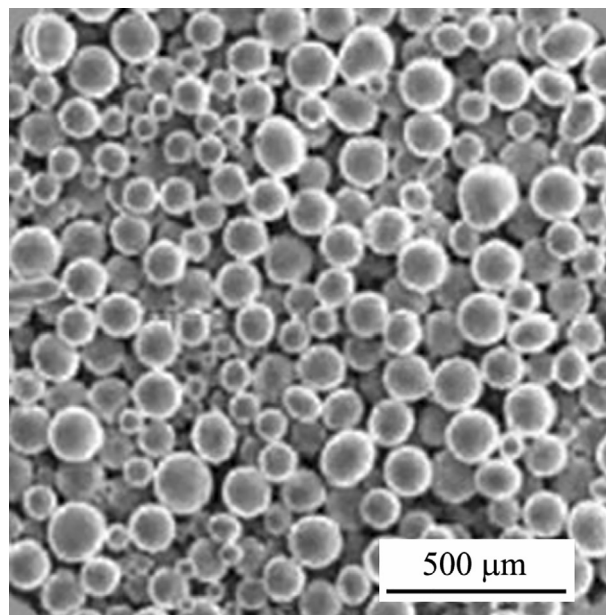


FIG. 18.7 SEM photomicrograph of a gas-atomized prealloyed Ti-6Al-4V. (From Froes FH. *Titanium powder metallurgy: developments and opportunities in a sector poised for growth*. *Powder Metall Rev* 2013; 2(4): 29–42.)

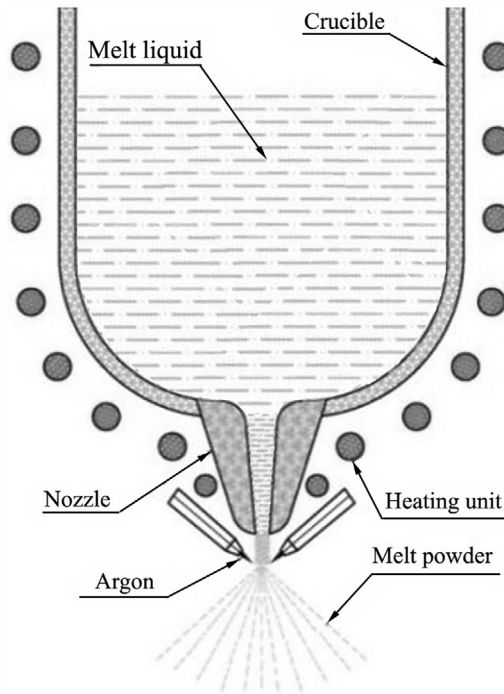


FIG. 18.8 Schematic of gas atomization process by experimental apparatus.

master alloy chips or granules is used [12]. Sintered MIM compacts using such Ti-6Al-4V alloy powder show that the tensile strength ranges from 810 to 880 MPa, and the tensile elongation is around 10%.

Vacuum induction gas atomizing (VIGA) is one of the gas atomization techniques to produce spherical titanium powders, including pure titanium, γ -TiAl, and titanium alloys. However, powders produced by VIGA commonly contain pores, which would be translated into the MIM and AM final products. In the work [13], the VIGA process was performed with high-purity argon gas. A schematic of the gas atomization process by experimental apparatus is shown in Fig. 18.8. The gas-atomized Ti-6Al-4V powders were mechanically sieved into two batches with various particle size ranges: ≥ 250 and ≤ 47 μm , respectively.

Fig. 18.9 shows the reconstructed X-ray computed tomography (CT) characteristic images of gas-atomized Ti-6Al-4V powders. As is shown in Fig. 18.9A, there are pores with various sizes in the coarse powder. The biggest pore size is about 55 μm while the smallest is below 8 μm (Fig. 18.9A). The authors [13] reported that the total porosity for the coarse powder is about 1.68% after calculation using the software of VG Studio MAX 2.2. Any observed pores in the small particle are presented in Fig. 18.9B.

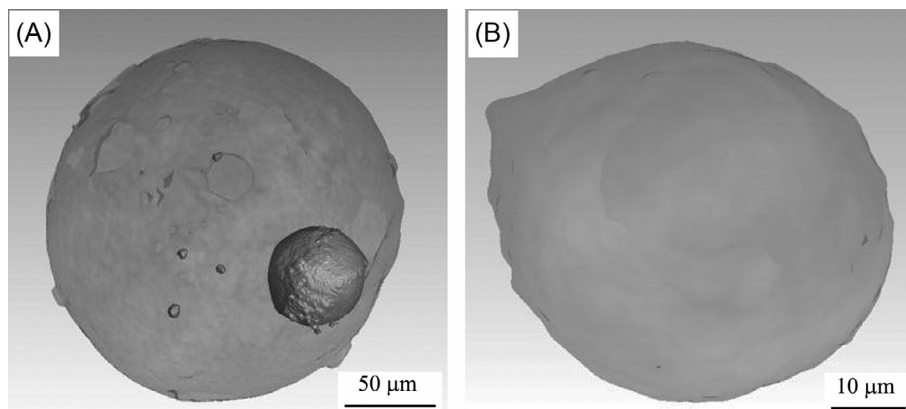


FIG. 18.9 Reconstructed X-ray computed tomography (CT) images of gas-atomized Ti-6Al-4V powders with the particle size of (A) ≥ 250 μm and (B) ≤ 47 μm ; the navy-gray spheres in (A) indicate the pore defects in the coarse powder.

For more information, including fundamentals of gas atomization, see [Chapter 4](#).

Numerous companies produce vacuum induction gas atomizing equipment. ALD Vacuum Technologies GmbH (Germany), ADMA Products Inc. (United States), SIMUWU Shanghai Gehand Vacuum Technology (China), SLT Zhuzhou Shuang Ling Technology Co. (China), etc., manufacture the vacuum induction gas atomization equipment in a wide capacity range. Thus, SLT Company produces such facilities of 1–1000 kg per batch capacity by vacuum pressure 0.05 Pa min. ALD Dynatech Furnace Pvt. Ltd. (India) manufactures VIGA from laboratory scale (1–8 L crucible volume) through pilot production (10–50 L crucible volume) up to large-scale atomization systems (with 300 L crucible volume). [Fig. 18.10](#) shows a large-scale inert gas atomization system with a melting crucible maximum capacity of two tons, equipped with a double-door melting furnace chamber and a gas recycling system to recover the inert gas.

ELECTROLYSIS OF TITANIUM COMPOUNDS

In the 1960s and 1970s, great attention was paid all over the world to titanium production by electrolysis of titanium halogenides (mainly, chlorides and fluortantalates) and titanium oxides (TiO_2 , Ti_2O_3 , etc.). However, the obtained electrolytic powder had less purity than magnesium reduction powder, so it did not get commercial use. The studies were carried out by using chloride electrolytes based on chlorides of alkaline and alkaline-earth metals (KCl-NaCl , KCl-NaCl-MgCl_2). Difficulties of TiCl_4 and TiO_2 electrolysis (main kinds of Ti raw material) are connected with their low solubility in chloride solutions (<1%). At the same time, it was possible to manufacture fine titanium powders on the cathode at low current yields due to special construction designs for TiCl_4 (bubbling) and TiO_2 (hashing) disintegration. The use of fluoride electrolytes allowed increasing the solubility of TiCl_4 and TiO_2 in the melt; however, in this case all the hardware-technological circuits of the electrolysis became very complicated and economically unprofitable.

A two-level method of electrolysis is considered as one of the prospective electrolysis techniques. In this method, at the beginning, the reduction of titanium tetrachloride in the melt to the tri- and bivalent condition at a low current density is carried out. Then the current density is raised, and titanium is precipitated on the cathode [\[14\]](#).

The new electrodeoxidation (EDO) titanium powder process is currently being developed and is based on a technique invented at Cambridge University known as the FFC Cambridge Process [\[15\]](#). The EDO process uses molten salt electrolysis to convert metal oxide into the pure metal. Unlike conventional electrowinning, in the EDO process the metal oxide is not dissolved in the electrolyte but is instead sintered into a ceramic material that then becomes the cathode of the electrolytic cell. During electrolysis, oxygen is extracted from the cathode and given off through the molten calcium chloride electrolyte on the anode. As the anode is made of graphite, this is consumed in the process and mainly carbon monoxide is given off. At the end of the process cycle, the cathode is converted into oxygen-free metal.

The titanium alloys can also be made using the EDO process by means of adding oxides of the alloying additions to the initial oxide mixture used to make the cathode. These are mixed in the given quantities such that after reduction, the metals



FIG. 18.10 Vacuum induction gas atomizing (VIGA) large-scale system with melting crucible maximum capacity of 2 tons. (From ALD Dynatech Furnace Pvt. Ltd.)

are in the correct proportions for the alloy. A large range of standard titanium alloy has now been made in this way. The process gives near-spherical titanium powder in the size ranges 20–250 μm in diameter. The powder particles are porous with a slightly irregular surface. The flowability of the dry powder is between that of atomized and HDH powders. Such a particle shape is favorable to give good green strength in processes such as pressing, sintering, or MIM.

FFS-Cambridge Process

In this process, solid titanium dioxide is cathodically reduced to titanium metal or titanium alloys in a molten salt (CaCl_2) (Fig. 18.11) [16]. The FFC process was developed by Fray, Farthing, and Chen at the University of Cambridge. It was realized by placing a metal oxide electrode in a molten salt bath heated to between 1073 and 1273 K and passing a current between it and an anode. At the interface between the current collector, the salt, and the metal oxide, the oxide ions leave the metal oxide and are carried across the salt to the anode, where they evolve as a gas.

The titanium dioxide is mixed with water to form a paste. It is then dried and pushed through an extrusion die to make the honeycomb or tablets. They are then fitted onto a wire, where they form the cathode in an electrolyte of molten salt with a graphite anode. Removal of a small amount of oxygen from the electrically insulating rutile phase converts it into the highly conducting Magnelli phase (TiO_{2-x}). Continued electrolysis removes oxygen from the cathode, where it dissolves in the electrolyte and is then removed as oxygen, carbon monoxide, or carbon dioxide at the anode. At a voltage of 2.5–3.3 V, no calcium is deposited. Process time is between 24 and 48 h, with resulting oxygen levels below 1000 ppm and nitrogen of 5–20 ppm. Longer processing times allow lower oxygen levels. Reduction of several oxides has allowed production of Ti-6Al-4V [17].

MER Process

The MER process uses a composite anode of TiO_2 , a reducing agent, and an electrolyte mixed with fused halides (Fig. 18.12). The composite anode approach containing stoichiometric carbon with the reduced TiO_2 is scaled to a daily output of 227 kg. The oxygen level of the batch products is below 500 ppm wt [18]. Predictions are for titanium production at a lower cost than the conventional Kroll process. Related background of the anode technology, applied to magnesium and aluminum, is provided in a set of expired patents [19–21]. Schematically, the process is shown in Fig. 18.12. The TiO_2 or rutile powders are mixed with carbonaceous material and binder, molded into electrode form, and heat-treated to form a composite anode. Ilmenite could be used for iron-containing alloys if the other impurities could be tolerated. The composite anode contains a reduced TiO_2 compound as $\text{Ti}_x\text{O}_y\text{-C}$. The Ti^{3+} ions are released into the electrolyte, further reduced, and deposited as titanium solid on the cathode. A CO/CO_2 mixture in bubble form is separated out at the anode. The titanium can be deposited as powder, flakes, or a solid deposit. Powder with particle size from 1 to 125 μm has been produced. The mean particle size can be controlled by process conditions.

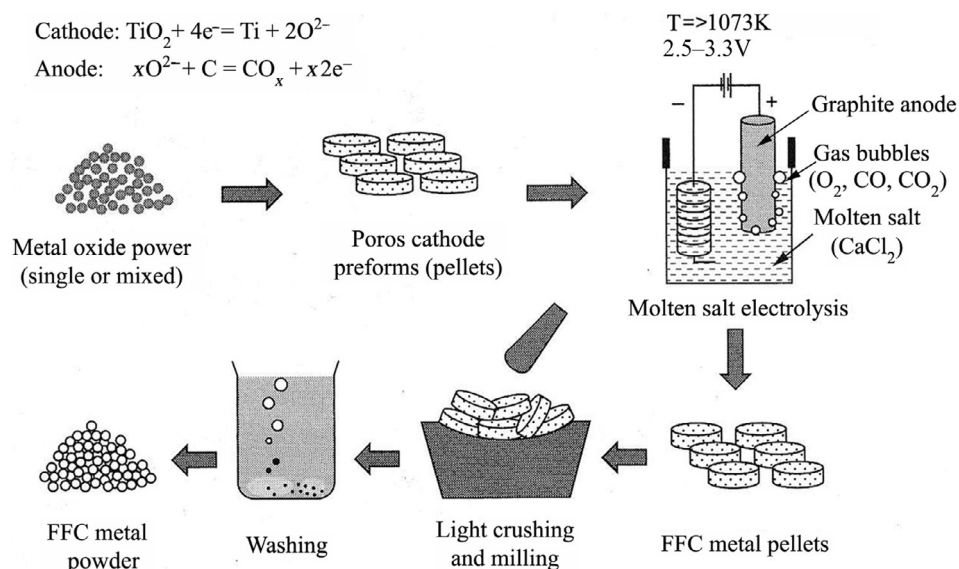
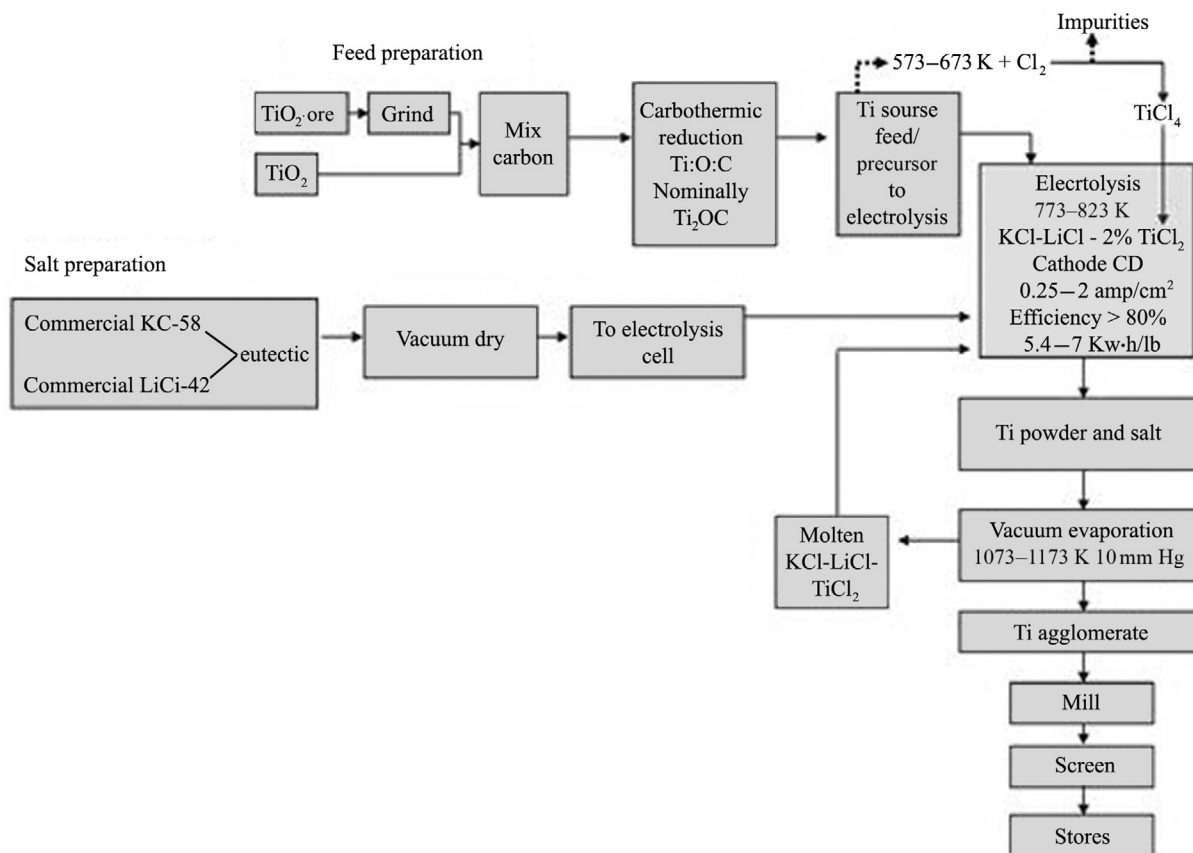


FIG. 18.11 The FCC Cambridge process.

FIG. 18.12 MER overall process to produce titanium powders from a TiO_2 .

Chinuka Process

This process was invented at Cambridge University in 2005 [16]. It allows impure metal oxides to be reduced and refined to high-purity metals in a single process. The process (Fig. 18.13) is very similar to the MER process and has a main advantage that the broad range of titanium dioxide can be used. Compared to the FFC Cambridge process, a eutectic of $\text{NaCl}:\text{KCl}$ is used as the electrolyte instead of CaCl_2 , which reduces the operating temperature of the cell by approximately 200 K . The Chinuka process offers a number of specific advantages over existing technologies, including comparatively low operating temperatures; a cheap, water-soluble, nontoxic electrolyte; low power consumption; and the ability to directly treat impure ores.

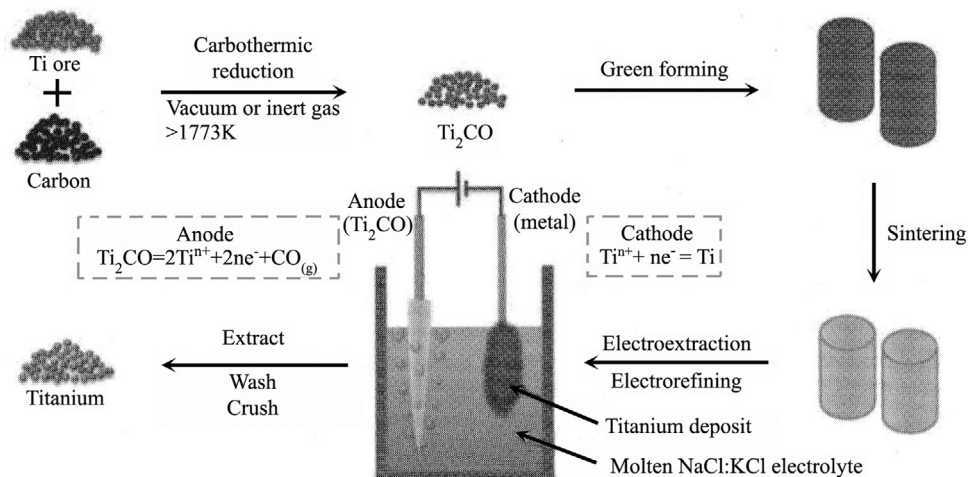


FIG. 18.13 Schematic illustration of the Chinuka process.

Electrolytic Refining of Titanium

This electrolysis process with a soluble anode was developed by the VAMI Institute [22]. The technological circuit of the electrolytic refining process is shown in Fig. 18.14 [12].

By anode dissolution of titanium metal, utilizing waste products or alloys, and cathode sedimentation of crystalline titanium, a titanium powder characterized by low content of impurities (Fe, Si, C, O, N, etc.) is produced. There are conditions permitting us to obtain powders alloyed with aluminum, chromium, vanadium, manganese, zirconium, and other metals.

Raw materials for anodes include substandard titanium sponge, alloys based on titanium, and conducting products of titanium raw material reduction (oxycarbides, oxycarbonitrides, etc.). The electrolyte is the molten NaCl, or NaCl + KCl, or NaCl + KCl + MgCl₂. The preparation of the electrolyte consists in the tetrachloride TiCl₄ reduction in molten NaCl and KCl using titanium scrap or sodium. Titanium ions are reduced to metal stepwise on the cathode; on the anode titanium is oxidized up to Ti²⁺ and Ti³⁺. An average degree of oxidation (valency of titanium ions) in the electrolyte is 2.2–2.3. The process of titanium refining was carried out in electrolyzers of VAMI design (Fig. 18.15).

The electrolyzer is a two-cell device with a common heating furnace. Each cell consists of a retort with bulk anode containers and a cathode chamber hermetically connected with the retort. Both cells are electrically connected in parallel. The cathode chamber is supplied with mechanisms for moving the cathode and for cutting off the deposit as well as the device for additional loading of the anode material, which consists of bunkers with feeders and discharge spouts with shutters. The device for unloading the anode material in the form of a mobile tray with a folding bottom and the metal receiver is mounted on the retort. The cathode is freely suspended from a top bridge with a shear device, which is connected with the elevating screw mechanism.

During electrolysis, the cathode is immersed in the electrolyte; a current supplying unit placed in the chamber performs the current feed to it. For deposit removal, the cathode is raised to the extreme top position and then is stopped; the top bridge moves downward on the cathode and cuts off the deposit, which falls into the tray. Then the top bridge is raised, the cathode is released and restores into the retort, and the cathode current circuit is closed. Impurities not dissolved at the anode are collected in the anode slurry and are periodically removed from the electrolyzer by overspill through a taphole. During the whole cycle of processing metal waste products, starting from the loading of the initial raw material and finishing with the unloading of the end product, the leak tightness of the electrolyzer under a small excess pressure of the inert gas is controlled.

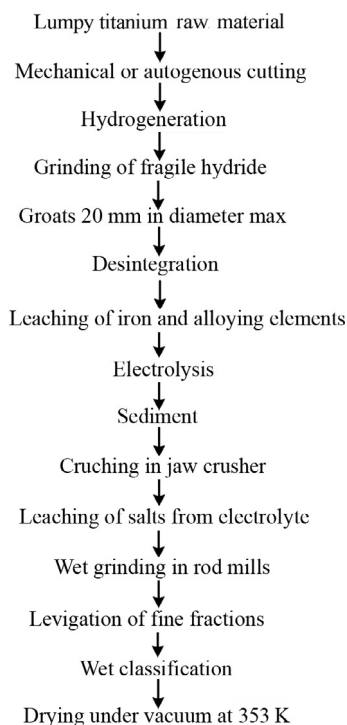
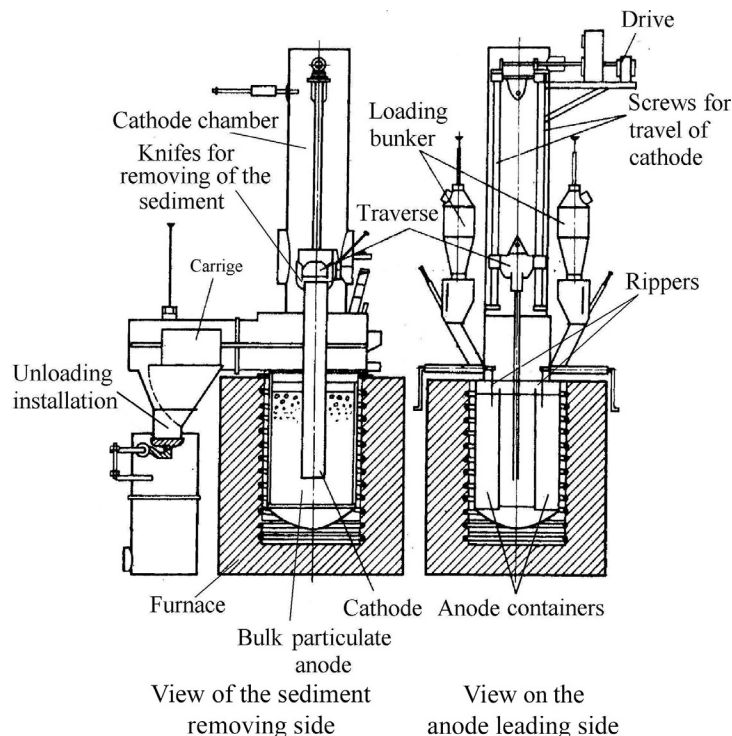


FIG. 18.14 Technological scheme of titanium powder production by the method of electrolysis with soluble anode (electrorefining).

FIG. 18.15 VAMI electrolyzer with a bulk anode for producing titanium powder.



The current strength in the electrolyzers of VAMI is 10–15 kA. The capacity of the electrolyzer for titanium refining is 120–180 kg/day (of metal). When using melt based on the KCl–NaCl mix with a content of 2%–6% titanium and a temperature of 1073–1123 K, the cathode current density is maintained in the interval of 5–15 kA/m² while the anode current density is 4–8 times lower. Current yield counted as Ti²⁺ is 90%, and the specific charge of electric power is 10–11 kWh/kg. Titanium extraction under these conditions amounts to 95%–96% with the consumption of the electric power directly to electrolysis of 4.0–5.0 kWh/kg.

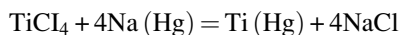
A high purity of powder purification is achieved with respect to iron, nickel, tin, molybdenum, silicon, nitrogen, oxygen, and carbon. The powders produced by electrolytic refining are more plastic than the powders produced by hydrogenation/dehydrogenation of the same titanium waste products. The method of electrolytic refining is realized at experimental industrial plants in Russia.

Physical and chemical characteristics and technological properties of the various kinds of titanium powders are shown in [Tables 18.1 and 18.2](#). In working with powders of chemically active metals, their transport, storage, and processing carry a danger of ignition and explosion. To lower the reactivity of powders, various methods of passivation are used ([Table 18.4](#)).

AMALGAM METALLURGY TECHNIQUE

In this process, titanium tetrachloride is reduced by sodium amalgam obtained by electrolysis with a mercury cathode and a graphite anode ([Fig. 18.16](#)).

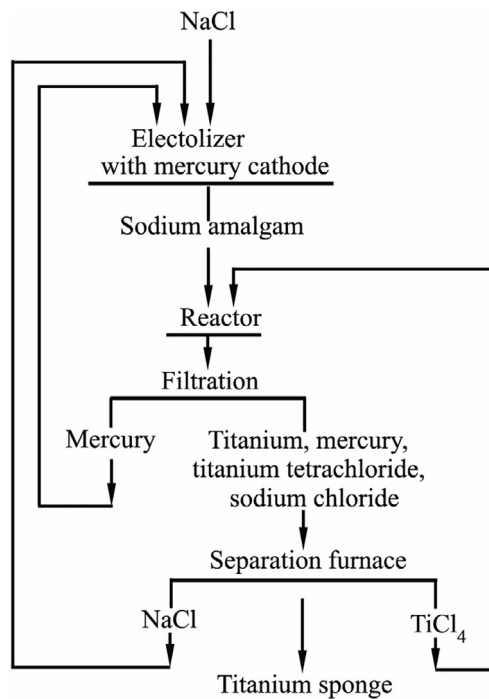
The process is carried out in a reactor in which the titanium tetrachloride is bubbled through the amalgam at a temperature of 361–377 K in an inert atmosphere (helium, argon, neon). The discharged chlorine is used for the chlorination of titanium dioxide, producing its tetrachloride:



The black powder formed is a mixture of titanium, mercury, sodium, and sodium and titanium chlorides. Mercury is removed in a vacuum furnace (vacuum of 1–10^{−2} Pa) or in inert atmosphere at 481–543 K. A mixture of titanium, its chlorides TiCl₂ and TiCl₃, and sodium chloride remains after distillation. NaCl is separated from the mixture in the inert atmosphere at 1088–1366 K. TiCl₄ reduction by a mixed sodium-potassium amalgam is carried out at 923 K. Low titanium chlorides decompose, and at a lowered pressure are removed. Sodium chloride is concentrated on walls and titanium

TABLE 18.4 Combustion Prevention for Titanium Powders

Manufacture	Branch of Industry	Technique of Powder Passivation
Production of refractory compositions and materials on their base	Nonferrous metallurgy, chemical and instrumental	Processing in acid solutions; high-temperature surface saturation; additives of aluminum oxide, nitrogen or carbon to argon plasma while dehydration
Production of friction materials by powder metallurgy	Engineering, petrochemical, etc.	Processing in aluminate solutions; processing by organosilicon composition liquids
Manufacturing of protective materials and coatings	Nonferrous metallurgy, food, and chemical	High-temperature oxidation (Combustion synthesis, consists in solid-phase reaction of titanium with titanium dioxide mixture). Use of epoxy resins for powder passivation
Production of sorbents and gas absorbers, and pyrotechnic compositions	Nuclear plant, electronics, pyrotechnics	Processing in acid solutions with addition of the spirits; surface metallization by cementation or electrochemical copper plating; insertion of metals into argon plasma during dehydration
Manufacturing of sintered constructional and porous materials	Nonferrous metallurgy, mechanical engineering	Processing in acids and solutions of halogenides; electrochemical and plasma-chemical surface metallization

**FIG. 18.16** Technological scheme of titanium powder production by the method of amalgam metallurgy.

crystals form a sponge. An advantage of the method is the production of particles of small size (1–10 μm) and the high purity of the titanium sponge.

The amalgam metallurgy fundamentals are discussed in [Chapter 8](#).

OTHER METHODS

TIRO Process

This process uses Kroll chemistry for manufacturing commercially pure titanium powder ([Table 18.5](#)) in a continuous fluidized-bed reactor ([Fig. 18.17](#)). This process also uses continuous vacuum distillation.

TABLE 18.5 Chemical Composition of TIRO Process Commercially Pure Titanium Powder

Element Content (wt%)							
Ti	O	N	C	Fe	Na	Cl	Other, total
>99.70	<0.15	<0.01	<0.01	<0.02	<0.02	<0.02	<0.01

Source: Barnes JE, Williams J, Blue CA, Peter B. Titanium powder metallurgy—what's after Kroll process? In: International titanium association conference. Atlanta (GA, USA); 2012.

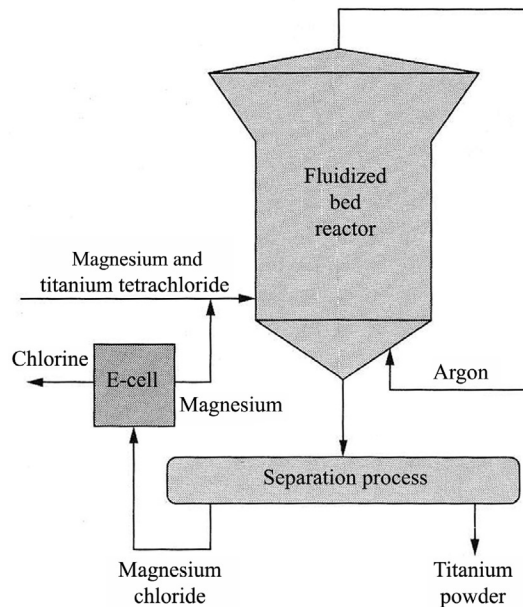


FIG. 18.17 Schematic of the TIRO process for manufacturing of commercially pure titanium powders. (From Barnes JE, Williams J, Blue CA, Peter B. *Titanium powder metallurgy – what's after Kroll process?* In: International titanium association conference. Atlanta (GA, USA); 2012.)

The advantages of the TIRO process include the following:

- Decreasing of specific energy consumption.
- Significant reduction of Kroll capital requirement.
- Low labor, high output, and continuity of the technological process.

This process developed by the Commonwealth Scientific and Industrial Research Organization (Australia) has been used for manufacturing thin sheets by direct powder rolling, extrusions, and cold-sprayed complex shapes.

Method of Disproportionation of Lower Titanium Halogenides

Low titanium halogenides (TiCl_2 , TiCl_3) disproportionate with the formation of metal powder under certain conditions of atmosphere, temperature, and pressure. The process consists of transferring low titanium halogenides of scrap metal or ores to titanium tetrachloride (TiCl_4) at high temperatures. During the process, the formed low chlorides are condensed in the colder zone of the furnace. On heating, they disproportionate with the formation of highly dispersed titanium powder and titanium chloride, which is returned to the beginning of the process. The produced powder is suitable for powder metallurgy.

CSIR Process

CSIR is the abbreviation of the Council for Scientific and Industrial Research that devised a continuous process to produce titanium powder directly by the metallothermic reduction of TiCl_4 in molten salt, named the CSIR-Ti process. The authors

[23] described the experiments for this process. The initial continuous stirred tank reactor has the capacity to produce 400–500 g of titanium powder per experiment. The reactor was loaded with 3–4 kg of an alkali metal chloride salt. Because the salt can be very hygroscopic, it was predried at 423 K for >6 h. The reactor was placed in a furnace and heated to reaction temperature above the melting point of the relevant salt, after which the reducing alkali metal and titanium tetrachloride were added. Following an experiment, the reactor was allowed to cool to room temperature prior to opening. The solidified mass of salt was removed by aqueous leaching to recover the metal powder produced. Because a slight stoichiometric excess of reducing metal was added, caution was taken during leaching because alkali and alkali earth metals react intensively with water. The reactor headspace was flushed with argon, followed by the slow addition of leach liquor to limit the risk of fire and explosion. Following leaching, the titanium powder product was collected. The metal powder was washed repeatedly to remove as much residual salt as possible.

Chemical Synthesis of Nanosized Titanium Powders

A major concern with titanium-based materials is their high cost compared to the competing materials. This had led to investigations on various processes with potentially lower costs. One such route is the synthesis of nanosized titanium powders from ilmenite by chemical methods [24,25]. In this study, commercially available ilmenite and commercially available or LR grade TiO_2 of 99.5% purity was used for the synthesis of the titanium complexes. Titanium hydride was prepared by heating the titanium complex in a hydrogen atmosphere. The furnace containing titanium catecholate/gallate was evacuated to 10^{-6} Torr followed by flushing with argon, and then hydrogen gas. The process was repeated three times before heating the titanium complex in order to avoid the formation of unwanted products. The furnace was then heated to 873 K stepwise from room temperature and collected all by-products while hydrogen was passed through the furnace for another 3 h. The titanium hydride formed was cooled under an argon atmosphere and stored in the same atmosphere. The vacuum dehydrogenation was done for getting pure titanium powder of required size and purity. The length of dehydrogenation and control of temperature affected the size distribution of the powders. The process is a green route because all the reagents from the process can be recovered and reused.

Mechanical Alloying

The use of mechanical alloying to produce titanium-based alloys is relatively recent. The process is a simple technique to synthesize both equilibrium and nonequilibrium phases of materials starting with elemental powders where, generally, a mix of powders is processed in a high-energy ball mill [26]. Much of the work consists of the development of unique structural characteristics such as supersaturated solid solutions, metastable crystalline phases, amorphous phases, dispersion-hardened alloys, and nanostructured materials [26–28]. Mechanical alloying is also being studied as a technique for the manufacture of nanostructured intermetallic composites such as Ti-6Al-4V/SiC [29]. The mechanical alloying process was used to synthesize $\text{Ti}_{45}\text{Zr}_{38}\text{Ni}_{17}$ powders for plasma thermal spraying [30]. The elemental powders mechanically alloyed by a planetary ball mill are led to an amorphous state. This amorphous phase that primarily appeared in the thermally sprayed mechanically alloyed powders is transformed into the icosahedral quasicrystal phase after annealing at 828 K. The Vickers hardness for the quasicrystal layers was about 7 GPa.

APPLICATIONS

Titanium powders that have unique properties are used as alloying, refining, and modifying components in the production of metals and alloys in ferrous and nonferrous metallurgy. They have high chemical resistance to many organic and inorganic aggressive matters at normal and high temperatures. The high-temperature strength, good plasticity at low density, and high corrosion resistance of titanium surpass many constructional and composite materials. The properties of titanium are realized most fully, and with high economic efficiency, by PM technologies. They are used as a basis of charges for production of refractory and composite materials and catalysts.

Titanium powders are readily formed and sintered; products so made possess high corrosion resistance, strength, and relative lightness. They are used for producing constructional details in mechanical engineering, ceramic-metal porous filters, welding and smelting electrodes, and gas absorbers in various designs of electronic vacuum devices. Titanium powders are widely used as fillers of anticorrosive coatings based on epoxy resin, in the production of titanium carbide and nitride, and in other fields of industry. One application field for powders of titanium and its alloys is the aviation industry.

Cases and bracelets for watches, surgical tools, prostheses, frames of lenses, weapon details, etc., are produced from titanium powders by the powder metallurgy processes. Porous titanium with 40% porosity and high mechanical properties with compression yield strength of 85 MPa have been manufactured by a plasma-spray method [31]. Takemoto et al. [31] reported that a complex interconnected porous structure, together with bioactivity effectively provided by chemical and thermal treatment, produced enhanced bone in-growth and apposition.

In recent years, second generation low-modulus near β and β -type titanium alloys have been developed for bioapplications in order to avoid the “stress shielding” effect caused by the modulus mismatch between the implant and the bone [32,33]. In the work [34], the electrochemical properties and biocompatibility of injection-molded Ti-10Nb-10Zr powder were studied. A feedstock containing 55 vol% powder loading was molded in a standard mold. The binders were completely removed from molded components by solvent and thermal debinding at different temperatures. The debinded samples were sintered at 1773 K for 3 h at high vacuum. The sintered density, ultimate tensile strength, elongation, and hardness were 99%, 738 MPa, 14%, and 111.6 HRB, respectively. The authors [34] consider the Ti-10Nb-10Zr alloys due to their high strength and good bioactivity as promising for application as biomedical materials. Higher yield strength was achieved [35] by metal injection molding of Ti-13Nb-13Zr parts using a feedstock based on elemental powders. As-sintered material showed a density of 96.5% of theoretical density, a modulus of elasticity of about 85 GPa, and a yield strength of 842 MPa, but of low <1% elongation. Ti-29Nb-13Ta-4.6Zr alloy, usually called TNTZ alloy, specimens were made by the MIM process [36]. The injection-molded compacts were sintered in vacuum at various temperatures between 1423 and 1523 K for various holding times between 14.4 and 28.8 ks. The microstructure of the sintered alloy mainly consisted of β phase. The tensile strength and elongation of the alloy increased with elevating sintering temperature, and were around 650 MPa and 3% at a sintering temperature of 1523 K, respectively. The elasticity of the alloy was around 60 GPa, and was similar to that of the wrought alloy. The laser-engineered net shaping (LENS) technique was used for the preparation of biomedical Ti-6Al-7Nb alloy [37]. The surface of Ti-Al-Nb samples obtained by the LENS system was modified by electrochemical treatment. The anodization of titanium alloy was performed in glycol-based and glycerol-based 0.3 M ammonium fluoride solutions with the addition of 2 vol% of water. It was reported that the used electrolytes are able to form TiO_2 with nanoporous/nanotubular morphology on the surface of the titanium alloy with Nb content and that a stable oxide layer of thickness 10–11 μm was obtained by anodizing the alloy in a glycol-based solution.

Until recent years, commercialization of titanium implantable components via MIM was an academic endeavor and not feasible due to its inherent processing difficulties [38,39]. The technological commercialization and process capability of titanium MIM to ensure consistent chemistry and mechanical properties is reviewed in article [40].

Porous elements are applied as filters in various designs of cartridge filters, as aerators in devices of regeneration and flotation saturation, and as devices of biological purification of sewage [22]. They are used as filtering elements in drainage devices as well as as gas absorbers, fire resistors, dehumidifiers, and oil separators. Elements can be welded by argon-arc, contact, and diffusion welding and can be processed using metal-cutting machine tools.

Despite its high cost in comparison with traditional permeable materials, titanium elements are used with high economic efficiency due to high strength, exceptional corrosion resistance, biological inertness, and low density. Also, filters can easily be regenerated, do not adversely affect the solutions being filtered, reduce losses, and allow the pressure and rate of filtration or aeration to be increased.

Depending on the particle size of the initial powder and the parameters of pressing and sintering, permeable materials from titanium powders have a basic pore diameter between 25 and 200 μm , a porosity of 30%–50%, and the minimal factor of permeability for various groups of products ranges from 1.0 to 2.5 mm/s. Porous titanium sheets with well-defined thickness in the range of 150–600 μm and porosities in the range of 8%–53% were produced by tape casting of commercial titanium powders [41]. Gas-atomized and HDH powders with particle sizes below 45 μm were used as starting materials. Currently, the best compromise was achieved if HDH powders were used as starting materials, sheet thickness was adjusted below 400 μm , and the sintering was done at 1273 K, which leads to wide open porosity around 30%. It has been demonstrated by electrochemical tests that these sheets can be used as a porous transport layer (PTL) and current collector in polymer electrolyte membrane (PEM) electrolysis cells.

Additive manufacturing (AM) is, in addition to widespread manufacturing of functional prototypes, increasingly deployed for manufacturing of titanium alloy components for a wide range of applications, including use in the medical, dental, aerospace, and industrial fields.

In recent years, AM has gone from being a rapid prototyping technology for plastic and metal components to being a manufacturing process considered for its flexibility, design freedom, and weight saving. The principle is that components are built directly, layer by layer, from three-dimensional CAD data. Metal printing technology is dominated by selected melting processes. However, emerging technologies, for example, binder jetting, are gaining interest and are today applied for prototyping as well as serial production of components. Basically, all types of powders with suitable morphology and

particle size can be applied to print green bodies. The subsequent sintering process is then optimized for each individual alloy. In the paper [42], the resulting properties from printing Ti-6Al-4V are presented. Digital metal technology, a two-step AM process, was applied for manufacturing Ti-6Al-4V components. Layer-wise precision binder jetting on a powder bed is followed by curing to develop the component's green strength. Subsequently, nonbonded powder is removed and component-sintered to develop density and strength. Process parameters are optimized for the alloy and physical properties examined in relation to material standard ISO 22068. In the study [43], the effects of titanium powder particle size on the mechanical properties and microstructures of AM parts manufactured by selective laser melting (SLM) were investigated. It is reported that the tensile strengths and elongations of the tensile specimens produced from different size powders (with median diameter 19.6, 22, 23.3, and 34.3 μm) by SLM at a laser power of 150 W and a scan speed of 200 mm/s vary in the ranges 610–470 MPa and 27%–11%, respectively. The technique of permeable coating production using a selective laser sintering method is described [44]. The coatings were made from titanium spherical powders using a layer-by-layer process from a metallurgical three-dimensional printer. The model of formation of the liquid metal necks between the powder particles by laser radiation for the effect of multiple reflections is reported.

The main mass of fine and medium titanium powders is used in the production of refractory compounds, particulate-reinforced titanium matrix composites, and metal-polymer anticorrosive compositions. Using self-propagating high-temperature synthesis (SHS process), it is possible to produce titanium carbide containing 79.3%–79.6% of titanium, 18.12%–19.42% of combined carbon, and 0.18%–0.28% of free carbon with high economic efficiency. The best results are achieved when using titanium powders finer than 0.1 mm. Powders of coarse fractions are used mainly as briquettes between 20 and 100 mm in diameter and 20–30 mm in height in ferrous metallurgy. Thus, only a part of the low-grade spongy titanium, which has not been converted to powders of average and fine fractions, is used in ferrous metallurgy. Besides, due to the high density of briquetted titanium and the small size of briquettes, the waste in their use is approximately 1.5 times lower than in direct use of spongy titanium. Batching and mechanization of briquette insertion into the molten steel is easily carried out.

Titanium Powder Process Selection

The characteristics of different types of titanium powders that are either available or under development at present are shown in Table 18.6.

In the powder process selection [45], the following aspects are considered:

- Cost of powder per unit weight. The powder that will approach most closely the price of titanium sponge is most likely to be high on the list for process selection.
- Powder process adaptability to scale up for powder production and for meeting major customer and market demands.
- Process controllability and lot reproducibility of the same product quality over many production lots as deliverables over time.
- Ability to satisfy the final processed product specification requirements given specific alloy chemistry, required product form, and so forth. Especially critical in this regard is the ability of the supplier to keep the production process under control and to satisfy the customer specifications for interstitial constituents.
- Minimizing the lead time. Powder producers must aim at greatly reducing that lead time to final deliverables.

The price of titanium sponge is usually used as the criterion for price comparisons for powder-based supply chain products. The two typifying different titanium PM technologies, namely blended elemental mixing powders of titanium with master alloy powders (BE) and the prealloyed (PA) methods, not only produce compacts with different sets of properties, but also with two different price ranges. At present, the most costly powders are those manufactured by PREP, REP, and GA methods. These technologies deliver powders at a price upward of \$100 per kg. This is because PA spherical powders for high performance applications and near 100% densification require an expensive melt stock, ultraclean handling, and expensive compaction tools. Over the past decade, the emerging titanium powder technologies for the future of low-cost titanium production had been developed. The deciding factor of these new processes will be which one of them delivers titanium powder at a cost as close as possible to the current cost of the titanium sponge.

Titanium Powder-Based Materials Selection

Alloy selection is determined by the part design requirements and the design allowable for the intended application. The powder consolidation processing route is selected for the best balance of mechanical properties, including tensile strength, ductility (elongation), fracture toughness, and fatigue endurance limit as well as fatigue crack growth resistance.

TABLE 18.6 Characteristics of Different Types of Titanium Powders

Method/Process	Powder Type	Advantages	Conditions/Disadvantages
Sodium reduction method (Hunter process)	Elemental	High-quality titanium; excellent for cold press and sinter	Less economical regarding the Kroll process; limited availability
Magnesium reduction method (Kroll process)	Elemental	Lower cost; good compactability; readily available; low chloride	
Calcium hydride reduction process	Elemental	Lower cost; readily available	
Hydrogenation/dehydrogenation (HDH) process	Prealloyed	Readily available	High cost; fair compactability
Armstrong process	Elemental and prealloyed	Continuous process; moderate cost; compactable	Processibility/quality; production scale up
Plasma-rotating electrode process (PREP)	Prealloyed	High purity	High cost; not cold compactable
Inert gas atomization	Elemental and prealloyed	High purity; available	Not cold compactable
Mechanical alloying	Elemental	Simple technique; low cost	Powder contamination
MER process	Elemental and prealloyed	TBD	Developmental
FES-Cambridge process	Elemental and prealloyed	Available	Heightened oxygen; production scaling up
Chinuka process	Elemental and prealloyed	Operating temperature reduced; low power consumption	Developmental
Electrolytic refining process	Elemental	High purity	Pilot plant
TIRO process	Elemental	Low energy consumption; continuous	Commercialization planning stage
CSIP process	Elemental	Simple technique	Pilot plant
Method of disproportionation of lower titanium halogenides	Elemental	TBD	Developmental
Chemical synthesis of nano size titanium powders	Elemental and prealloyed	Low cost	Developmental

TBD, To Be Determined.

Modified from Froes FH. Titanium powder metallurgy: review—part 1. *Adv Mater Process*; 2012; 16–22.

Selection of the desired combination of processing steps must also be kept in mind in determining the final product cost. For preliminary guidance in making specific choices of powder-based titanium material selection, the designer is referred to the analysis and concluding remarks summarized previously in the current article for the tables covering processing-microstructure-property trends for given processing step sequences.

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